

Armin Salmasi

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WORK EXPERIENCE

Software Engineer & Model Developer

2023 - Present

Thermo-Calc Software AB

- Designed and developed scalable Python backend services, APIs, and ML models for material property predictions in a production environment.
- Built and maintained internal tools and platforms to streamline data workflows and ML model deployments for cross-functional engineering teams.
- Managed CI/CD pipelines, automated testing, and build environments using Jenkins, ensuring high reliability for production systems.

Key Projects:

- **GenAI Toolchain:** Engineered LLM-based code assistants and a RAG pipeline to autonomously generate Python scripts, enhancing developer experience.
- **ML Platform & Workflow:** Spearheaded end-to-end development and orchestration of ML workloads, enabling self-service and automated execution of complex computational tasks.
- **Data Infrastructure:** Developed robust ETL pipelines and deployed scalable NoSQL database solutions for efficient storage and retrieval of scientific data.

R&D Simulation Specialist

2023

Epiroc Drilling Tools AB

- Analyzed large-scale manufacturing datasets to optimize production workflows and build data-driven design capabilities.
- Collaborated with cross-functional teams to translate domain-specific challenges into scalable computational and statistical models.

Key Projects:

- Data-driven design optimization using advanced statistical analysis and predictive modeling.
- Simulation of complex physical dynamics under high-pressure conditions.

Research Intern (Master Thesis)

2016

Sandvik Coromant AB

- Applied **Density Functional Theory (DFT)** calculations to screen high-entropy alloy candidates.
- Developed diffusion simulation models to predict gradient formation during sintering.

TECHNICAL SKILLS

Languages	Python (Advanced), Java, SQL, Bash Script
Platform & DevOps	GCP, Docker, CI/CD, Git, Jenkins
ML & AI	PyTorch, TensorFlow, MLFlow, LLMs (RAG, Prompt Eng.), LangChain, Scikit-learn
Data & Backend	APIs, Distributed Systems, ETL Pipelines, MongoDB, MySQL, Pandas
HPC	MPI, OpenMP, Distributed Computing, SLURM
Tools	Linux, macOS, VSCode, IntelliJ, PyCharm, VIM

EDUCATION

Ph.D., Engineering Materials Science

2016 – 2022

KTH Royal Institute of Technology

Focus: ICME, Mass Transport Simulation, and Numerical Modeling. Served as a Teaching Assistant for multiple advanced master's level courses.

M.Sc., Materials Science

2016

KTH Royal Institute of Technology

Focus: Simulation of Gradient Formation using Thermodynamic and Kinetic modeling.

M.Sc., Nano Materials

2011

Materials & Energy Research Center

Focus: Thin Film Deposition and Characterization.

B.S., Extractive Metallurgy

2007

Amirkabir University of Technology

CERTIFICATIONS & TRAINING

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|---|------|--|------|
| ▪ Supervised ML: Regression & Classification , DeepLearning.AI | 2023 | ▪ Neural Networks & Deep Learning , DeepLearning.AI | 2020 |
| ▪ Good Manufacturing Practice (GMP) , Svensk Medicin AB | 2022 | ▪ Python: Functions & Dictionaries , U. Michigan | 2019 |
| ▪ Software Tools: ML to Phase Diagrams , MGF | 2022 | ▪ Python Basics , U. Michigan | 2019 |
| ▪ Practical Deep Learning , ENCCS | 2021 | ▪ The Unix Workbench , Johns Hopkins University | 2019 |
| ▪ CodeRefinery (Software Development) , NeIC | 2020 | ▪ CAMD Summer School (DFT) , DTU Physics | 2018 |
| ▪ Structuring Machine Learning Projects , DeepLearning.AI | 2020 | ▪ CSC HPC Summer School , Solvalla, Finland | 2017 |
| ▪ Improving Deep Neural Networks , DeepLearning.AI | 2020 | ▪ PDC HPC Summer School , KTH, Sweden | 2017 |
| | | ▪ RACIRI Summer School on Materials Design , Rugen | 2015 |

PUBLICATIONS

 Please refer to my [Google Scholar Profile](#) for a complete and up-to-date list of my publications.